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Project Title:	High-Precision Microwave Spectroscopy of Long-Lived Circular-State Rydberg Atoms in Microgravity		
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	Objective: The system of fundamental constants (the speed of light, Planck's constant, the fine structure constant, the Rydberg constant, etc.) is important because it directly reflects our current understanding of the foundations of nature. The objective of the present project is to measure one of these constants, the Rydberg constant, at an uncertainty level in the range of a few parts in 10^12 (the current uncertainty stands at 5 parts in 10^12). It is planned to use circular Rydberg atoms, which are very large, highly excited atoms with doughnut-shaped electron distributions that do not penetrate into the atomic nucleus. As a result, these atoms are not susceptible to perturbations caused by electron wavefunction overlap with the nuclear charge density and quantum-electrodynamics (QED) corrections. It is this property that makes a measurement of the Rydberg constant with circular Rydberg atoms particularly useful and potentially highly competitive, in comparison with already established methods. Due to the absence of critical QED		

Task Description:	corrections and nuclear penetration, a measurement of the Rydberg constant with circular Rydberg atoms would help resolve a recently reported inconsistency, where it was concluded that "... either the Rydberg constant has to be shifted by -110kHz/c (4.9 standard deviations), or the calculations of the QED [quantum-electro-dynamic] effects in atomic hydrogen or muonic hydrogen atoms are insufficient." [R. Pohl et al., Nature 466, 213 (2010)]. Method: Rb atoms are cooled to below 1microK and laser-excited into a low-angular-momentum Rydberg level with principal quantum number n on the order of 40. Using an adiabatic atomic-state transformation method, the Rydberg atoms are then transferred into the circular state |n,n1=0,n2=0,m=n-1> (m, n1, and n2 are the magnetic and parabolic quantum numbers). An optical lattice (wavelength 1064nm or 532nm on the ground, 850nm at CAL) is adiabatically ramped on. The Rydberg atoms are trapped in the lattice via the ponderomotive interaction of the Rydberg electron with the lattice light. During the hold time of the atom cloud in the lattice, the electric-quadrupole transition into the state |n+2,n1=1,n2=1,m=n-1> is driven by amplitude-modulating the optical lattice at the transition frequency. The transition is probed using state-selective electric-field ionization. The measured transition frequency allows one to extract the Rydberg constant. Key advantages of the method are that the transition is free of first-order Stark and Zeeman shifts, that QED, nuclear-penetration, hyperfine and other shifts are small, and that the optical lattice causes only moderate trap-induced transition shifts. Theoretical work includes the modeling of the circular-state preparation using multi-level adiabatic passage and of the circular-state transition dynamics in a shallow, microwave-modulated optical lattice, in which center-of-mass wave-packet expansion and tunneling must be considered. Relevance to NASA and relevance of microgravity: High-precision measurement of fundamental constants represents an important part of contemporary atomic-physics research. The proposed high-precision measurement of the Rydberg constant would add a new component to the Cold Atom Laboratory (CAL) research portfolio. To achieve optimal spectroscopic performance it is critical to reduce systematic lattice-induced transition shifts and related shifts. Since the most critical shifts diminish with decreasing optical-lattice depth, the lattices must be chosen as shallow as possible. Microgravity conditions are important because they enable the use of shallow lattices without the atoms falling out. The microgravity environment provided by the CAL, including its 850nm optical-lattice laser, offers an ideal platform for the project to reach its ultimate uncertainty goal. The CAL setup may also provide an extended period of measurement time to acquire good statistics in the high-precision spectra.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	High-precision spectroscopy and metrology impacts fields such as time measurement and navigation. Benefits to the general public include atomic clocks and the GPS system. The project “High-Precision Microwave Spectroscopy of Long-Lived Circular-State Rydberg Atoms in Microgravity” involves the development of a fundamentally new type of spectroscopy, which has been successfully demonstrated during the first year of the grant. The future development of laser systems to hone this innovative technique may lead to benefits in other disciplines, such as chemistry, biology, and engineering. The research involves graduate students, undergraduate students, and post-docs, who are being prepared for future jobs in academia, government, and industry. Having a capable scientific workforce is essential for progress and competitiveness in science and technology, as well as for the national economy. The main goal of the project, which is to measure the Rydberg constant with circular Rydberg atoms, will help resolve the “proton radius puzzle.” The puzzle has arisen from inconsistencies in measurements on low-lying electronic and muonic states of hydrogen, which involve the electromagnetic interaction, QED, the proton structure, and QCD. The project will therefore serve as an indirect test for QED and the structure of the proton. With Rydberg atoms, the Rydberg constant will be tested at electron energies in the range of milli-electronvolts. This regime is rather different from the energy regime of previous measurements (eVs to keVs).
	The project “High-Precision Microwave Spectroscopy of Long-Lived Circular-State Rydberg Atoms in Microgravity” deals with measuring the Rydberg constant and other atomic constants, such as polarizabilities, using long-lived circular Rydberg atoms. A Rydberg atom is an atom in a high-lying energy level. Rydberg atoms are very large in size (up to micrometers, which is huge for individual atoms) and have long lifetimes, which is a desired feature because it is conducive to the achievement of high precision in measuring the frequencies of transitions between Rydberg states. The precise measurement of the Rydberg constant to within 5 parts in 10^12 (the present goal) and of other atomic constants relies on that capability. Circular Rydberg atoms are particularly well suited for this purpose, because they are longer-lived than regular atoms, their Rydberg electron distribution does not penetrate into the atomic nucleus (and hence the details of the nuclear change distribution are irrelevant), and other perturbations (quantum-electromagnetic corrections, for instance) are also not important. A careful analysis, based in part on previous work by researchers at the MIT (Group Leader D. Kleppner et al. [1,2]), the ENS (Group Leader S. Haroche et al. [3, 4, 5, 6]) and the MPQ in Garching (Group Leader H. Walther et al.[7]), has shown that the atomic transitions that are most promising for the task are electric quadrupole transitions between circular Rydberg states. Specifically, a transition type is required that has zero transition-frequency corrections due to weak electric or magnetic fields that are typically present in experimental chambers. The large size of the Rydberg atoms allows one to employ a fundamentally new method of spectroscopy. The method is based on harnessing a term known as the “A-square term” for the purpose of spectroscopy. The method requires one to have a modulated light field with a spatial modulation period that is on the same order as the diameter of the atom that is being probed. At the same time, the field must be periodically modulated in time, with a modulation frequency that matches the frequency of the atomic transition that is to be probed. Rydberg atoms in electro-optically modulated standing wave light fields satisfy both conditions. It must be stressed that (1) this method of making an atomic transition has not been done before, and (2) that the method is a critical part in the the project “High-Precision Microwave Spectroscopy of Long-Lived Circular-State Rydberg Atoms in Microgravity.” To that end, in the first year of the project the research team has tried and succeeded in demonstrating this new type of spectroscopy. The group is leading in Rydberg-atom trapping in magnetic fields and in optical lattices. An optical lattice is a laser-induced crate for atoms, made by standing waves of light, that researchers use to enhance spectroscopic resolution and to prepare the trapped atoms for quantum gates, spectroscopy, etc. The optical lattice is often compared with an egg carton that holds eggs in a way that is equivalent to how periodic laser trap holds cold atoms. The Raithel group has first characterized such traps and has now begun using them for high precision spectroscopy. The group’s effort to help solving the “proton radius puzzle” by measuring transitions between laser-trapped circular Rydberg atoms has previously been awarded with a National Institute of Standards and Technology (NIST) Precision Measurement Grant (2012-2015), and is now being continued within the Fundamental Physics Program of the NASA Cold Atom Lab.

Task Progress:	In 2014/2015 the group succeeded in measuring the 58S-59S quadrupole Rydberg transition by “A-square,” or “ponderomotive,” lattice modulation spectroscopy. The transition has a frequency of about 38GHz. This work, which was published in Nature Communications, is crucial for the present project. To reach high relative spectroscopic resolution, it is imperative to both decrease the spectroscopic linewidth (the frequency uncertainty to which transitions can be pinpointed) and to increase the absolute frequency of that transition. In the present case, the targeted numbers are 1Hz for the uncertainty of the transition, and 200GHz for the transition frequency itself (corresponding to a relative uncertainty of 5 parts in 10^12). Since no suitable methods exist to electro-optically modulate the optical lattice 200GHz exists, it is critical to find a workaround that allows one to increase the accessible frequency range from 40GHz, the approximate limit of current electro-optic modulation technology, to 200GHz. In 2015 the research team has succeeded in identifying such a workaround. Essentially, the nonlinearity of electro-optic modulation allows one to drive all odd overtones of the modulation frequency, without the need of increasing the optical-lattice laser power. Currently, the team has shown that transitions with frequencies up to 5 times the modulation frequency can be driven. The team has already demonstrated a transition at about 95GHz. The goal of about 200GHz is therefore within close reach. This work has been published in Physical Review Letters. In the last year, the team has designed a new cryogenic spectroscopic setup that is supposed to last through the entire performance period of about 4 more years. The setup features a three-dimensional optical lattice for Rydberg atom trapping and lattice modulation spectroscopy. It incorporates all ingredients necessary for preparing the above-mentioned circular Rydberg states, which will be required to reach the desired spectroscopic accuracy and precision. The circular-state production method follows pioneering work by Group Leader D. Kleppner et al. at MIT (Massachusetts of Technology) [8]. The setup is ready for cryogenic operation at 4 Kelvin, which will be required in order to suppress unwanted transitions and lifetime reductions due to black-body thermal radiation. The setup also includes the required laser-cooling infrastructure as well as control of electric and magnetic fields within the spectroscopic volume. The setup has proceeded well into the construction phase. The measurement of the Rydberg constant at a precision of 5 parts in 10^12 necessitates improved knowledge of the ionic core polarizability of the utilized rubidium atoms. The core polarizability accounts for the fact that the atomic core (the Rb+ ion in this case) becomes electrically polarized by the Rydberg electron that orbits the Rb+ ion. The core polarization leads to Rydberg level shifts on the order of 10kHz. A small, simplified spectroscopy chamber, assembled from spare parts and some new components, has been set up to perform high-precision microwave spectroscopy on states with angular momenta of up to about five. Transitions between such states are ideal to determine the core polarizability. The goal of this component of the work is to increase the number of significant digits for the core polarizability from presently three to five. The five significant digits will be sufficient to eventually extract the Rydberg constant at a precision of 5 parts in 10^12. The experimental work is accompanied by a theoretical effort conducted by V. Malinovsky. The present objective of the theory work is to model the spectroscopic line shapes of Rydberg transitions in an amplitude-modulated Rydberg-atom lattice in a manner that takes the quantization of the center-of-mass motion into account. This ability will be an important ingredient to determine the line center with 1Hz uncertainty. To take into account the quantization of motion of the Rydberg atoms in the optical lattice, a theoretical model is developed in which both ground and excited wave functions are subject to the periodic potentials, representing the lattice fields, while the coupling (the effective Rabi frequency) is also periodic as function of the translational coordinate. To obtain the spectrum of the excited-state population, the time-dependent Schroedinger equation will be solved in the momentum representation. References: [1] R.G. Hulet, D. 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Bibliography Type:	Description: (Last Updated: 02/24/2025)
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